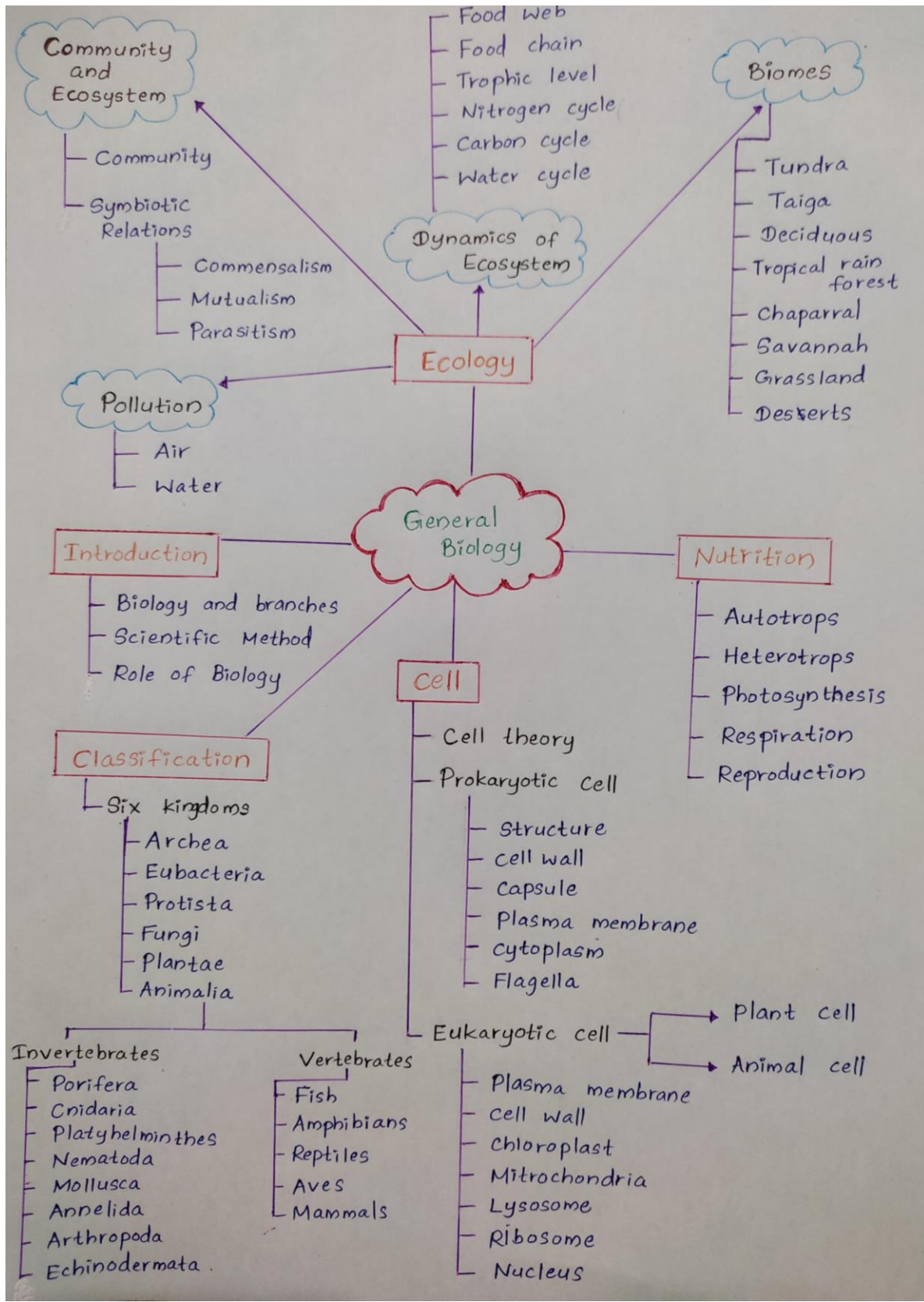


Ecology



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Communities and Ecosystem

Community - All organisms that live together in a place is called community.

Co evolution – The Plants, animals, protists, fungi, bacteria etc. live together and have changed and adjusted to one another continually over a period of millions of years. Such interactions are called co evolution.

Symbiotic Relationship: - Shown by two or more kinds of organisms, live in a place in permanent relationship.

Types of Symbiosis – Commensalism, Mutualism, Parasitism

Types of Symbiosis	Meaning	Example	Describe of Example
Commensalism	✓ Give benefits for one species ✓ Neither hurts nor helps the other ✓ Species are physically attached	Epiphytes and tree (Orchid and mango tree)	• Epiphytes that grow on the branches of other plants. • They benefit by having the substrate for growing. • But host plant is unharmed
Mutualism	✓ Organisms that live together by means of benefiting each other.	Ants and aphids	○ Aphids suck fluids from the phloem. Then they excrete through their anus. ○ Ants carry the aphids to new plants. Then consume the excreted materials of aphids. ○ So, aphids get new plant, ants get food
Parasitism	✓ Harmful to the prey organisms and beneficial to the parasite ✓ Parasites much smaller than the prey. ✓ Closely associated with host	Lice and man	▪ Lice lives in the human skull and get nutrients and habitat

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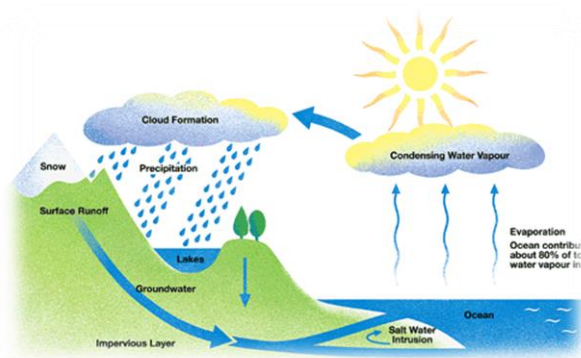
Dynamics of the Ecosystem

- ✓ An **ecosystem** is a community of living organisms interacting with each other and their non-living environment.
- ✓ Ecosystem makes up by: -
 - all living things (plants, animals and bacteria)
 - non-living things (the sun, rocks and soil etc.)
- ✓ Every ecosystem has regulated transfer of energy and controlled cycling of nutrients.
- ✓ All of the substances cycle through ecosystem in bio-geo-chemical cycles.

What is the bio-geo-chemical cycle?

- The chemical interactions (cycle) that exist between the atmosphere, hydrosphere, lithosphere and biosphere.

Water Cycle: -



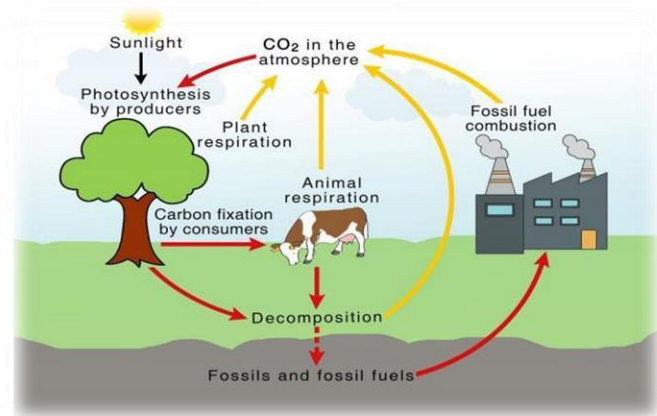
The water cycle is driven by energy from the sun. The water cycle is composed of many parts.

During part of the water cycle,

The sun heats up liquid water and changes it to a gas by the process of **evaporation**. Water that evaporates from oceans, lakes, rivers, and moist soil rises up into the atmosphere. The process of evaporation from plants is called transpiration. As water (in the form of gas) rises higher in the atmosphere, it starts to cool and become a liquid again. This process called **condensation**. When a large amount of water vapor condenses, it results in the formation of clouds. When the water in the clouds get too heavy, the water falls back to the earth. This is called **precipitation**. When rain falls on the land, some of the water is absorbed into the ground forming pockets of water called groundwater. Most groundwater eventually returns to the ocean. Other precipitation runs directly into streams or river. Water that collects in rivers, streams and oceans is called **runoff**.

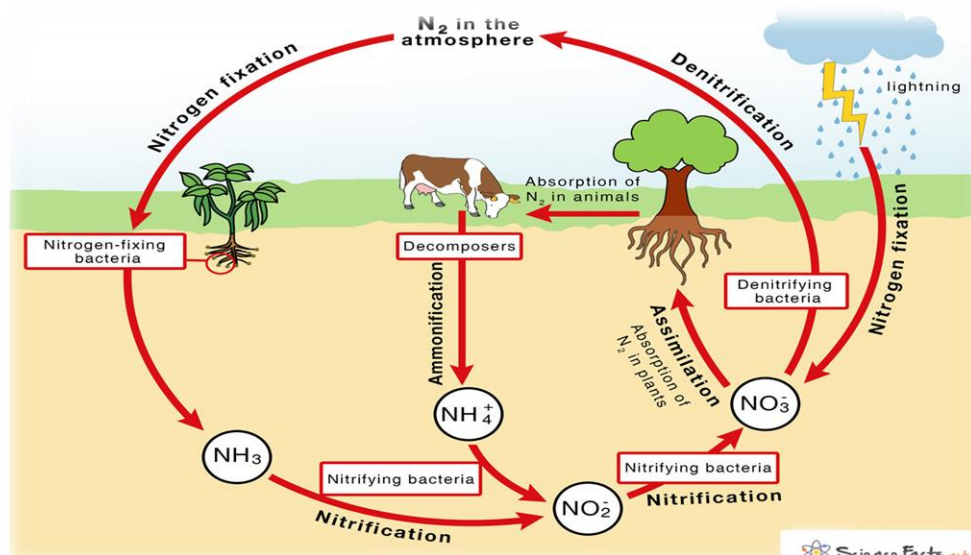
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Carbon Cycle: -



- ✓ During photosynthesis, plants, algae, and cyanobacteria remove CO_2 from the air fix, or incorporate it into complex organic compounds such as glucose.
- ✓ When organisms eat plants, they take in the carbon and some of it becomes part of their own bodies
- ✓ When organisms eat animals, they take in the carbon and some of it becomes part of their own bodies.
- ✓ They generate CO_2 from cellular respiration and exhale this CO_2 into the atmosphere.
- ✓ When animals and plants die, most of their bodies are decomposed and carbon atoms are returned to the atmosphere. (CO_2)
- ✓ Some are not decomposed fully and end up as deposits underground (oil, coal)
- ✓ Carbon moves from fossil fuel to the atmosphere when fuels are burned, when human burn fossil fuels to power factories, power plants and vehicles.
- ✓ Additional carbon is stored in the ocean; CO_2 passes into the water from the atmosphere. Plants and algae use this CO_2 for photosynthesis. Ocean contain earth's largest store of carbon.

Nitrogen Cycle: -



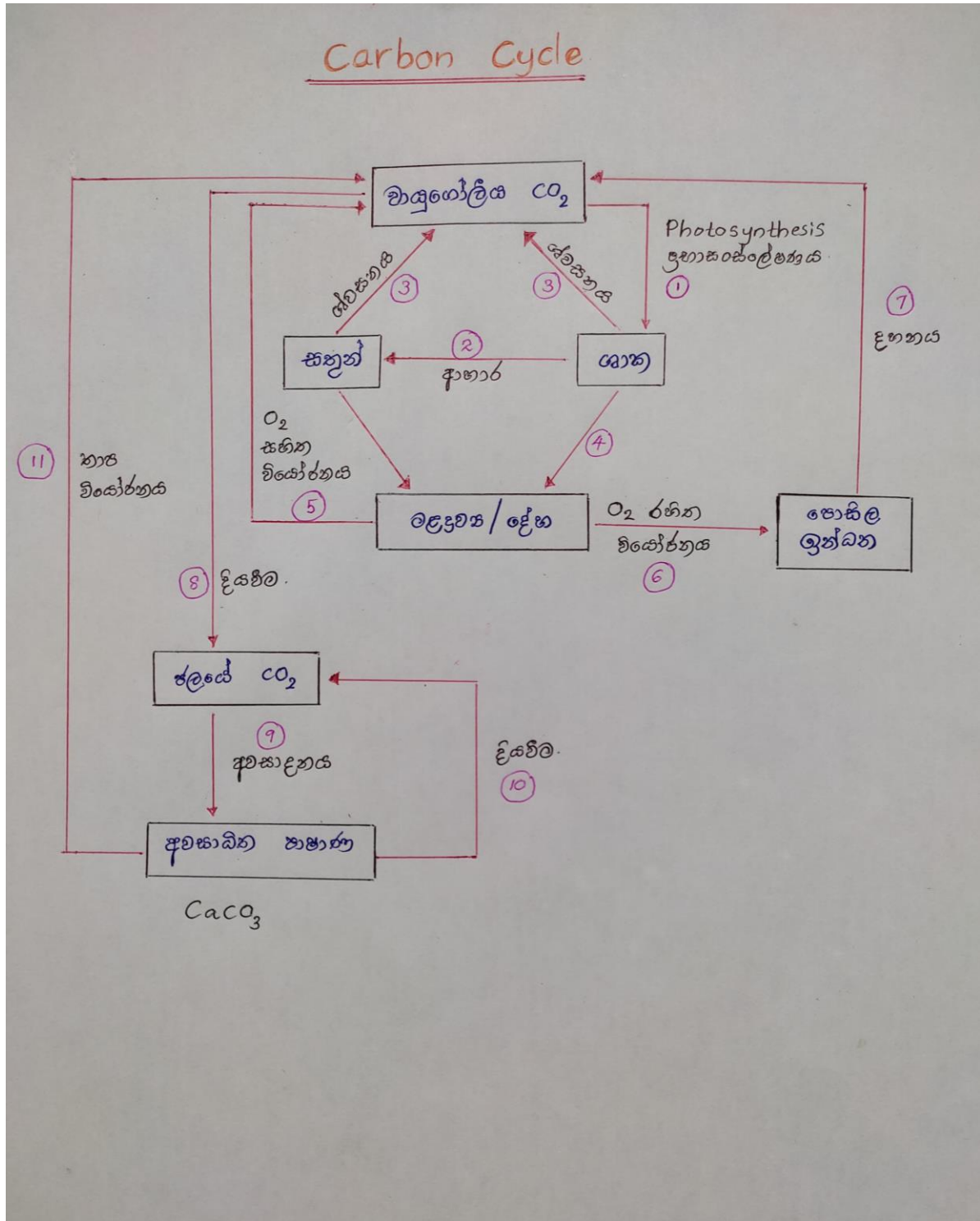
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- The most important part of the Nitrogen cycle is bacteria. Bacteria help the nitrogen change between states so it can be used. When nitrogen is absorbed by the soil, different bacteria help it to change states so it can be absorbed by plants. Animals then get their nitrogen from the plants.
- **Fixation** - Fixation is the first step in the process of making nitrogen usable by plants. Here bacteria change nitrogen into ammonium.
Fixation bacteria – *Rhizobium*, *Azotobacter*
- **Nitrification** - This is the process by which ammonium gets changed into nitrates by bacteria. Nitrates are what the plants can then absorb.
Nitrification bacteria – *Nitrococos*, *Nitrobactor*
- **Assimilation** - This is how plants get nitrogen. They absorb nitrates from the soil into their roots. Then the nitrogen gets used in amino acids, nucleic acids, and chlorophyll.
- **Ammonification** - This is part of the decaying process. When a plant or animal dies, decomposers like fungi and bacteria turn the nitrogen back into ammonium so it can reenter the nitrogen cycle.
- **Denitrification** - Extra nitrogen in the soil gets put back out into the air. There are special bacteria that perform this task as well.
Denitrifying bacteria - *Pseudomonas*

Significances of bio-geo-chemical Cycles

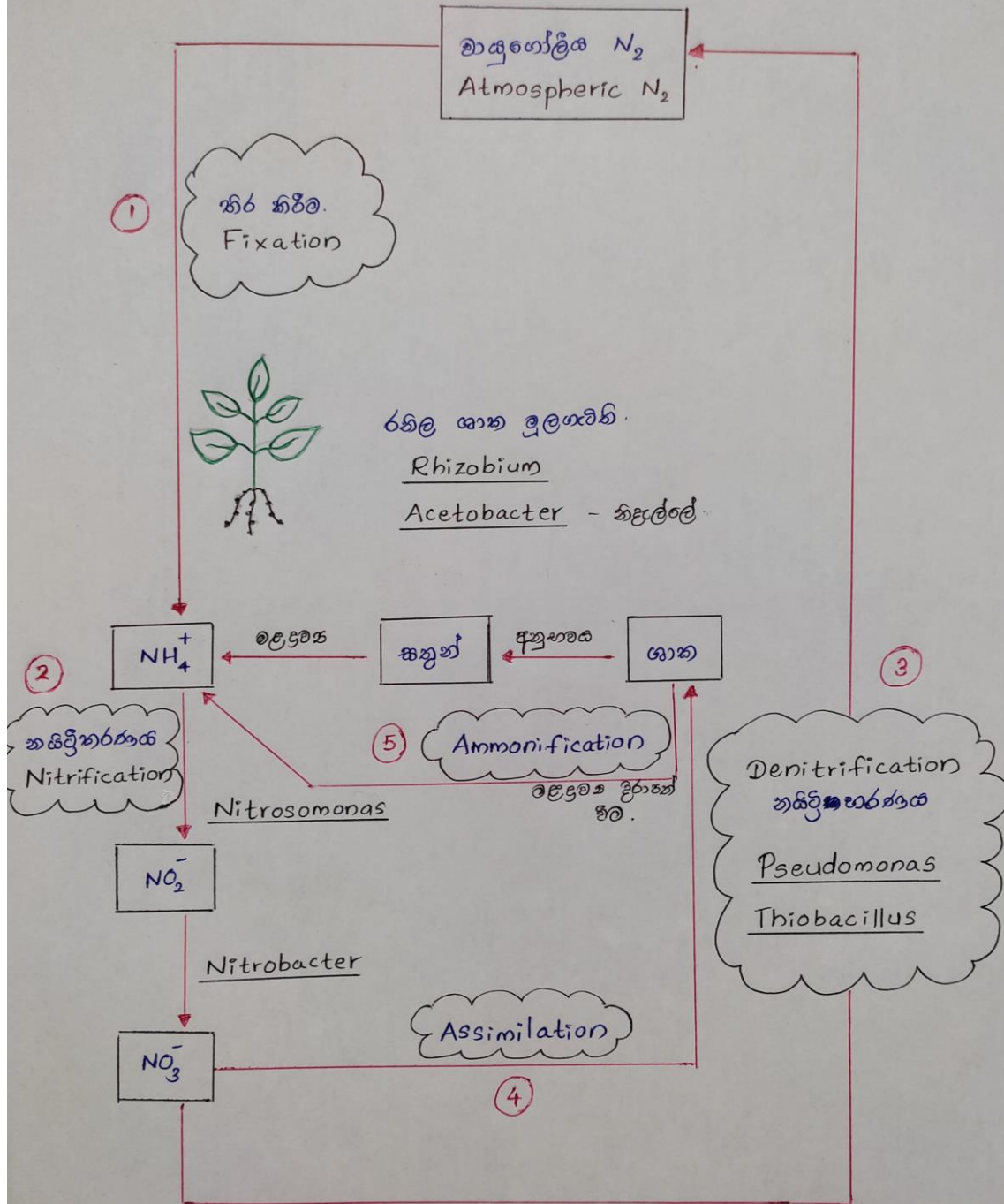
- ❖ Assist in the functioning of ecosystem.
- ❖ Facilitate the strong of elements.
- ❖ Enable the transformation of nutrients from one to another form.
- ❖ Enable the transfer of molecules from one locality to another.

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Nitrogen cycle

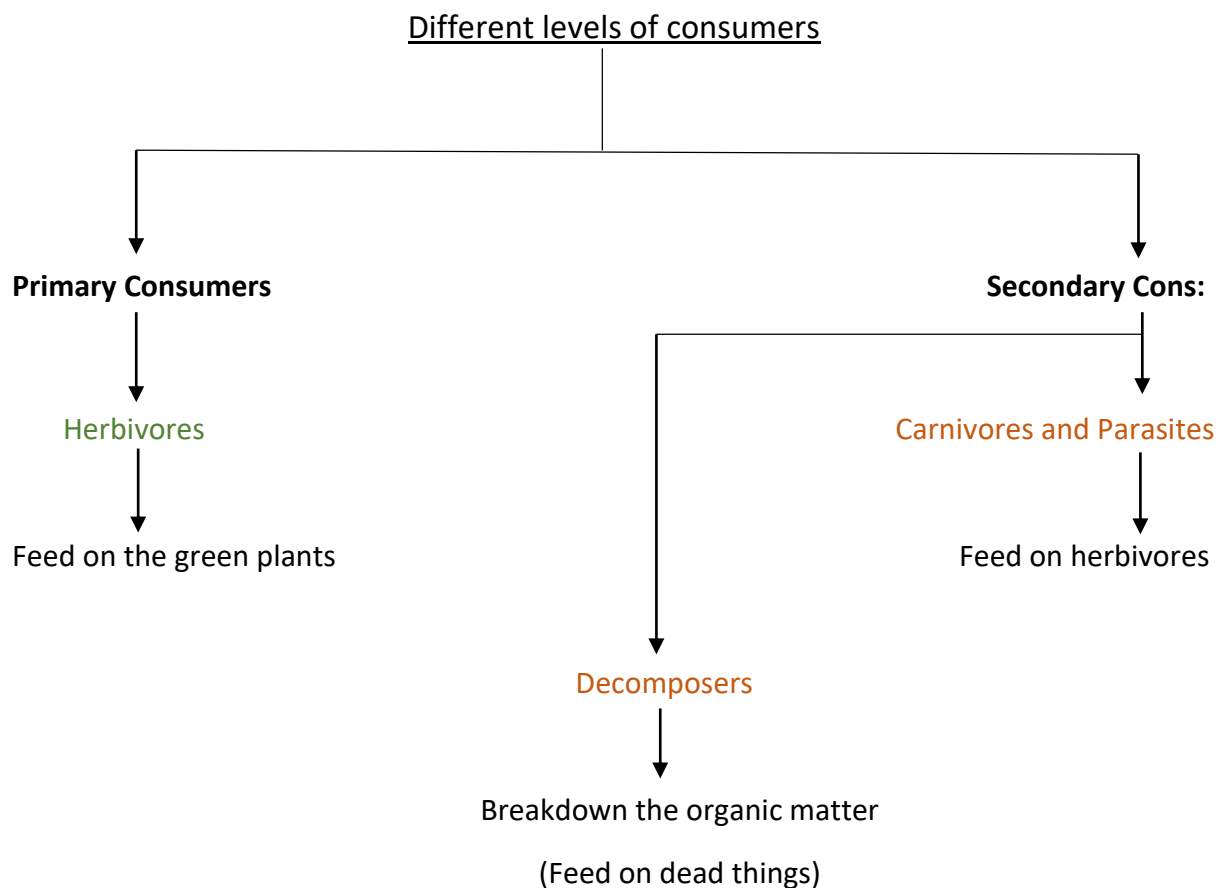


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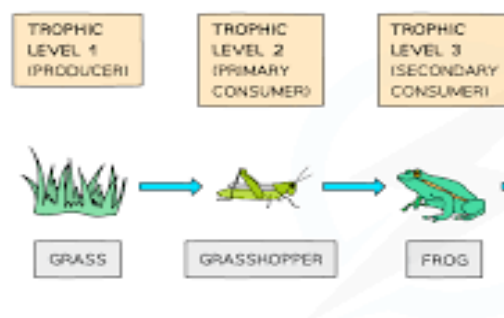
Trophic Levels

Autotrophs – able to capture light energy and manufacture their own food. Examples for plants, Algae and some bacteria. They are **primary producers**.

Heterotrophs – Obtain organic molecules that have been synthesized by autotrophs. Examples for animals, fungi and most other organisms. They are consumers.



- All of these categories (Autotrophs, Primary and Secondary consumers...etc.) are occurred in any ecosystem.
- They represent different “TROPHIC LEVELS”
- (Examples for trophic levels: - Plants, Primary and Secondary Consumers.etc.)



Ecology

What is the meaning of food chain?

- ✓ Organisms from each trophic level feed on one another make up a series called food chain.
- ✓ A food chain transfers food energy from plants to an animal and then to another animal. Most food chains start with green plants.

What is the meaning of food web?

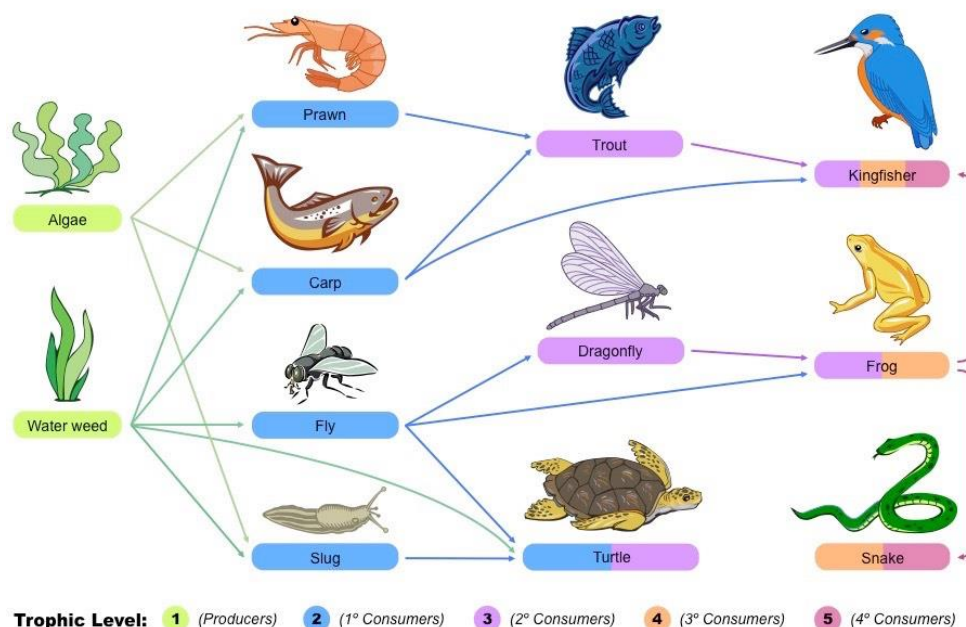
- ✓ A community of organisms where there are several overlapping food chains with different pathways for the flow of food energy in the ecosystem.

Significance of Food Chain: -

- The knowledge of food chain helps in understanding the feeding relationship as well as the interaction between organism and ecosystem.
- It also helps in understanding the mechanism of energy flow and circulation of matter in the ecosystem.
- It also helps to understand the movement of toxic substances and problems associated with biological magnification in the ecosystem.

Significance of Food web: -

- Food webs are important tools in understanding that plants are the foundation of all ecosystems and food chains, sustaining life by providing nourishment and oxygen needed for survival and reproduction.
- The food web provides stability to the ecosystem



Ecology

BIOMES

Characteristics Features of Biomes: -

- Biomes is the largest terrestrial community or the largest ecological unit.
- It has specific kinds of plants and animals.
- It has distinct climate conditions.
- Biomes are distributed on earth as broad belts.

The Major Biomes

Name	Features	Dominant Plants	Precipitation	Distribution
Tundra	• Poor Light and extremely low temperature • Snowfall is scanty • Sparse vegetation	Dwarf willow trees, lichens and mosses	25 cm ↓	Asia, Europe, North America
Taiga	• Average temperature 6°C in winter 20°C in summer • Evergreen trees	Pinus, Spruce	10cm-35cm	North America, Europe, Asia
Deciduous forest	• Occurred in temperate region	Broadleaved hardwood trees	75cm-150cm	N.C Europe, eastern Asia, Eastern United states
Tropical rain forest	• Occurred in equatorial regions • Warm and humid climate • Luxuriant and varied vegetation	Ceiba, Kapok, Teak	140cm ↑	Central and south America, Western India, Southeast Asia
Chaparral	• Limited Winter rain followed by drought	Broad-leaved evergreen shrubs, bushes and small trees usually less than 2.5m tall.		Mediterranean sea, Pacific coast of south America, South Australia
Tropical Savannah	• Warm climate	Grasses, Baobab	100cm-150cm	South America, Africa, Asia and Australia
Grasslands	• Short and tall grasses found	Buffalo grass, Blue grama	25cm-75cm	Prairies Canada, USA Pampas South America Steps Europe, Asia Veldts Africa
Desserts	• Extremely low rainfall • High evaporation • Hardy grasses found	Saguaro castus, Joshua tree	25cm ↓	North Africa, South Western USA

Ecology

MCQ: -

1. Animals that help breakdown the dead organisms that fall to the ground are_____
 - a. Producers
 - b. Consumers
 - c. Decomposers
2. A connected group of food chains is a_____
 - a. Habitat
 - b. Food web
 - c. Ecosystem
3. Animals that eat other animals are_____
 - a. Biomes
 - b. Herbivores
 - c. Carnivores
4. What organisms make their own food with sunlight or chemical energy_____
 - a. Producers
 - b. Consumers
 - c. Decomposers
5. Which cycle most likely includes sunlight, photosynthesis and respiration,
 - a. Water cycle
 - b. Carbon cycle
 - c. Nitrogen cycle
6. Lightning, legumes and bacteria are part of the_____
 - a. Carbon cycle
 - b. Nitrogen cycle
 - c. Water cycle

Ecology

7. Through which process do living organisms release carbon into the atmosphere_____
- a. Photosynthesis
 - b. Cellular respiration
 - c. Combustion
8. The _____ is a land ecosystem that gets very little rain.
- a. Tundra
 - b. Desert
 - c. Grassland
9. Which biomes has the greatest diversity of living things,
- a. Desert
 - b. Taiga
 - c. Rainforest

Answers

1) c 2) b 3) c 4) a 5) b 6) b 7) b 8) b 9) c

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